NON-STABILIZATION OF CHOW SEMISTABLE PLURICANONICAL LIMITS FOR SMOOTH HYPERSURFACES

ARGUS OPEN RESEARCH PROJECT

Abstract. We record the theorem of Wang and Xu (2014) disproving the stabilization of Chow semistable r-canonical limits as $r \to \infty$ for degenerating one-parameter families of canonically polarized varieties, including smooth hypersurfaces of large degree in $\mathbb{P}^3$.

1. Main Refutation

Theorem 1.1 (Wang–Xu, 2014)**.** *There exist flat projective families of canonically polarized manifolds over a punctured curve (including degenerations of smooth surfaces of degree $d \gg 1$ in $\mathbb{P}^3$) whose asymptotic Chow semistable limits with respect to rK_X do not stabilize as $r \to \infty$ whenever the K-semistable limit differs from the asymptotic Chow limit.*

Date: August 27, 2026.

2020 *Mathematics Subject Classification.* Primary 14J10; Secondary 14L24, 32Q20.